DIFFERENT TYPES OF RESEARCH

NAME :- RAGYAM SINHA
ROLL :-13000123064
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BASIC VS. APPLIED RESEARCH (THE "WHY")

- Basic (Fundamental) Research: Driven by curiosity. Its goal is to expand knowledge and develop theories without immediate commercial use (e.g., studying how the brain processes light).
- Applied Research: Designed to solve practical, "real-world" problems (e.g., developing a more efficient solar cell). Visual Content: A side-by-side comparison of a scientist looking through a telescope (Basic) vs. an engineer fixing a wind turbine (Applied).

QUANTITATIVE RESEARCH (THE "HOW MANY")

- Focus: Numerical data, statistics, and objective measurements.
- Common Methods: Surveys, polls, and systematic experiments.
- Goal: To generalize results from a sample to a larger population. Visual Content: A clean dashboard featuring bar graphs, pie charts, and a calculator.

QUALITATIVE RESEARCH (THE "WHY & HOW")

- Focus: Exploring human behavior, experiences, and social patterns.
- Common Methods: In-depth interviews, focus groups, and case studies.
- Goal: To understand the meaning people attach to their experiences. Visual Content: A silhouette of two people talking with "thought bubbles" containing various icons (heart, house, lightbulb).

MIXED-METHODS RESEARCH (THE "FULL PICTURE")

- The Approach: Combining both quantitative and qualitative data to provide a holistic view.
- Example: Using a survey to find how many people like a product, then interviewing them to find out why. Visual Content: Two puzzle pieces—one with numbers and one with words—snapping together.

DESCRIPTIVE VS. ANALYTICAL RESEARCH (THE "WHAT" VS. "HOW")

- Descriptive: Fact-finding. It describes "what is" without influencing the variables (e.g., a census).
- Analytical: Uses existing information to critically evaluate and find causal relationships (e.g., why did the stock market crash in 2008?). Visual Content: A camera (representing a snapshot/Description) vs. a prism (breaking down light/Analysis).

LONGITUDINAL VS. CROSS-SECTIONAL (THE "TIME" FACTOR)

- Longitudinal: Studying the same subjects over a long period (years or decades).
- Cross-Sectional: A "snapshot" of different groups at one specific point in time. Visual Content: A timeline with many points (Longitudinal) vs. a single vertical slice through a group of people (Cross-sectional).

EXPERIMENTAL RESEARCH (THE "CAUSE & EFFECT")

- The Gold Standard: Manipulating one variable (independent) to see its effect on another (dependent).
- Setup: Often uses a "Control Group" and an "Experimental Group." Visual Content: A laboratory setting with test tubes and a checklist.

- Context: Commonly used in education and management.
- Process: A cycle of "Plan → Act → Observe → Reflect."
- Goal: To improve specific professional practices on the go. Visual Content: A circular arrow diagram representing the cycle of improvement.

ACTION RESEARCH (THE "IMPROVEMENT")

CONCLUSION & SUMMARY

- No single research type is "better"; the choice depends on your Research Question.
- Key Takeaway: Choose your method based on whether you need depth (Qualitative), breadth (Quantitative), or a solution (Applied). Visual Content: A "Library of Knowledge" or a diverse group of researchers collaborating.

THANK YOU